Practice workbook · Set up and explain the indicator practical comparing carbon dioxide in breath

Two-tube limewater or indicator apparatus with a mouthpiece: drawing and using it, naming the indicator, and explaining results for inspired, expired, rest or exercise air.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BREATH-CO2-INDICATOR-PRACTICAL

1. Question 5 · part a.ii · 3 marks

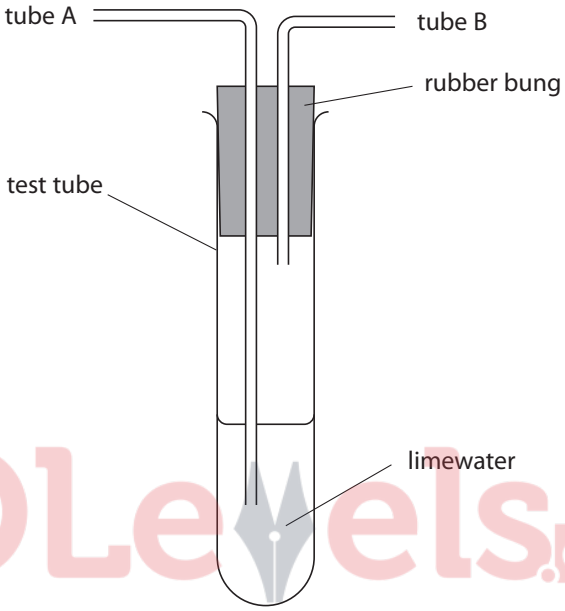
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Worked-example retry · Attempt part a.ii only.

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5 The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.



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(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

(4)

1. Question 5 · part a.ii · 3 marks

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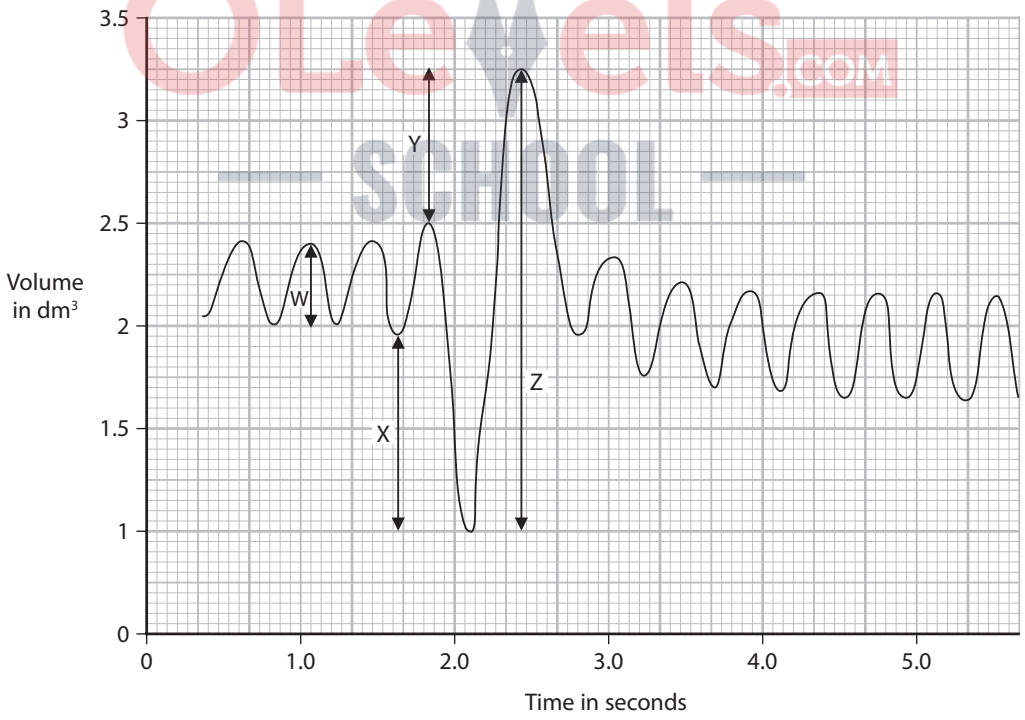
Worked-example retry · Attempt part a.ii only.

(ii) Explain the difference expected between the results.

(3)

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.



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1. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

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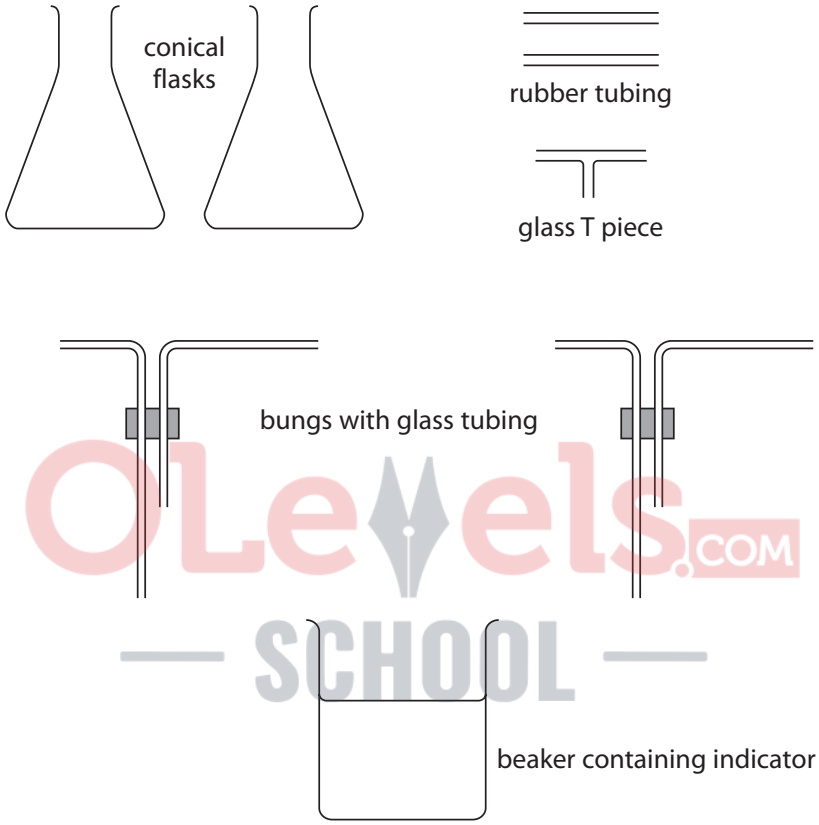
2. Question 1 · part a · 3 marks

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Further practice · Attempt part a only.

1 A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.



The apparatus is not drawn to scale.

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2. Question 1 · part a · 3 marks

June 2022 Paper 2R

Further practice · Attempt part a only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

(3)

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(b) Name an indicator that could be used in the investigation.

(1)

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(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

(4)

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2. Question 1 · part a · 3 marks · Working and answer

Working and answer

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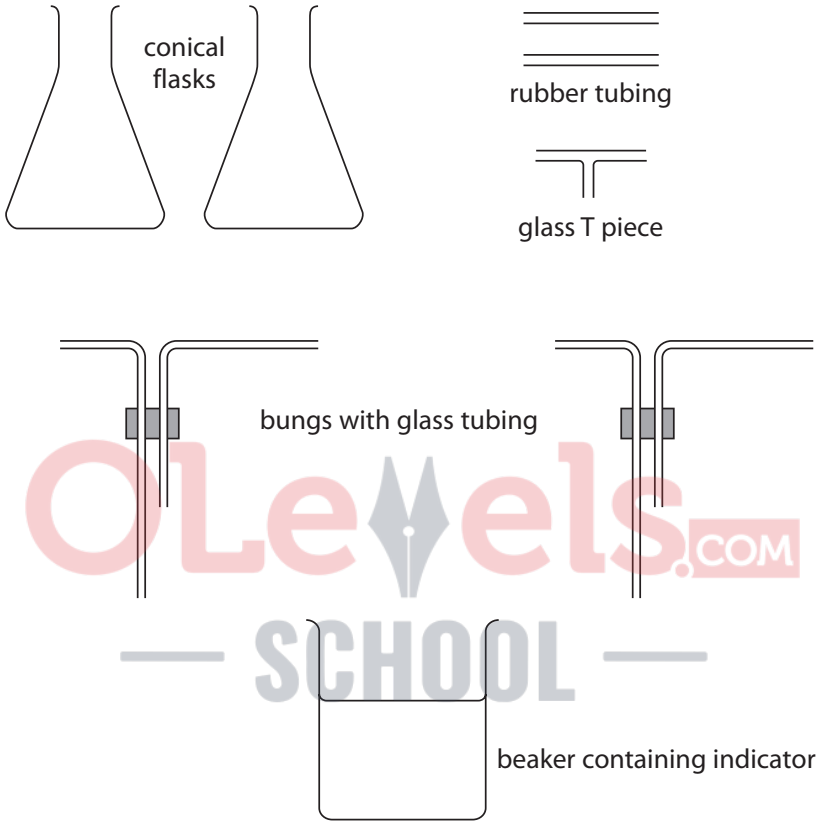
3. Question 1 · part b · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b only.

1 A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.



The apparatus is not drawn to scale.

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3. Question 1 · part b · 1 mark

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Further practice · Attempt part b only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

(3)

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(b) Name an indicator that could be used in the investigation.

(1)

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(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

(4)

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3. Question 1 · part b · 1 mark · Working and answer

Working and answer

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4. Question 2 · part b.i · 1 mark

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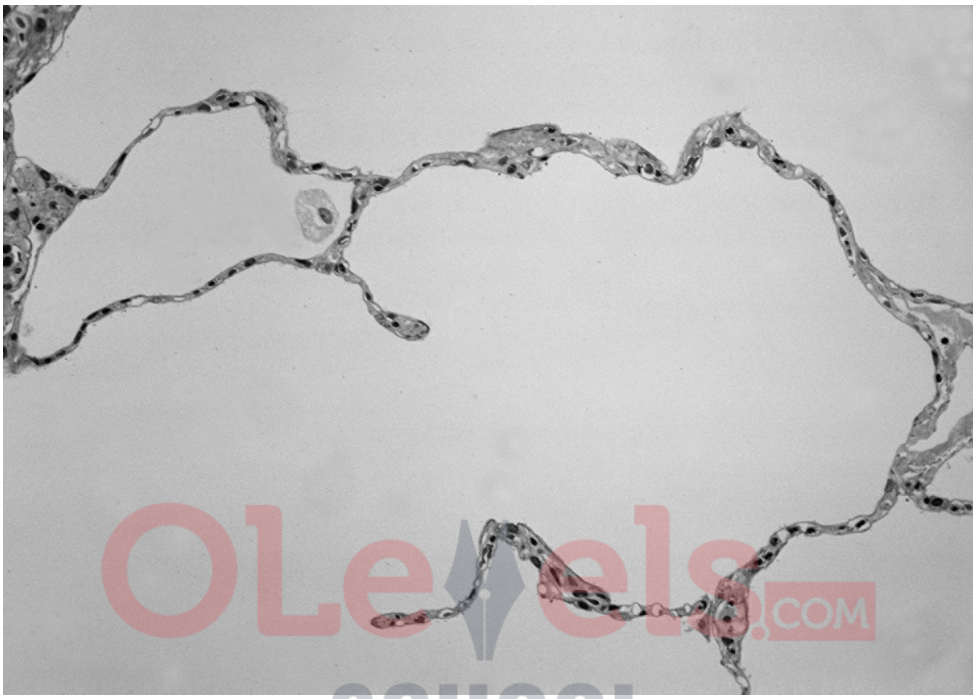
Further practice · Attempt part b.i only.

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2 The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

(2)

.....

.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

(2)

1 .....

.....

2 .....

.....

4. Question 2 · part b.i · 1 mark

June 2023 Paper 1

Further practice · Attempt part b.i only.

(b) (i) Which is the percentage of carbon dioxide in the atmosphere?

(1)

- ☒ **A** 0.01
- ☒ **B** 0.04
- ☒ **C** 0.10
- ☒ **D** 0.50

(ii) The table shows the results obtained when a person at rest breathes in air that contains different percentages of carbon dioxide.

Percentage (%) carbon dioxide in air	0.02	1.60	3.00	6.00
Tidal volume in cm^3	500	700	1200	2100
Rate of breathing in breaths per minute	14	15	16	28

Calculate the volume of air in dm^3 breathed in per minute when the person breathes air containing 3.00% carbon dioxide.

[1 dm^3 = 1000 cm^3]

(2)

volume of air = ... dm^3

(iii) The air a person breathes in contains 20% oxygen. The body uses 20% of this oxygen.

Calculate the volume of oxygen used per minute when the person breathes air containing 1.60% carbon dioxide.

(3)

volume of oxygen = ... cm^3

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P 7 2 5 8 5 A 0 4 2 0

4. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

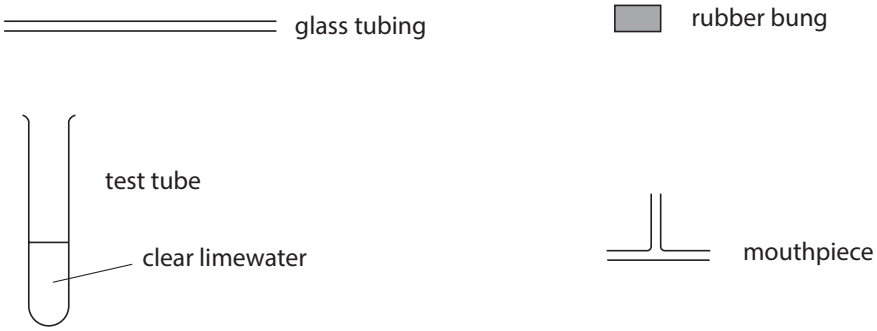
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5. Question 4 · part a.i · 3 marks

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Further practice · Attempt part a.i only.

4 (a) The diagram shows some scientific equipment.



(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

(3)

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5. Question 4 · part a.i · 3 marks

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Further practice · Attempt part a.i only.

(ii) State one safety precaution in this investigation.

(1)

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

Mean breathing rate in breaths per minute		
at rest	**after gentle exercise**	**after vigorous exercise**
14	19	50

(i) Give one control variable in this investigation.

(1)

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

(4)

(Total for Question 4 = 9 marks)

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5. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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6. Question 7 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

7 (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

(1)

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

(2)

$C_6H_{12}O_6$ + ... ⟶ ..... + ... CO_2

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

(2)

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P 7 4 4 9 5 A 0 2 0 2 8

6. Question 7 · part a.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.iii only.

(iv) Describe how a student could use this equipment and a timer to compare the rate of carbon dioxide production during exercise and at rest.

(4)

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6. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

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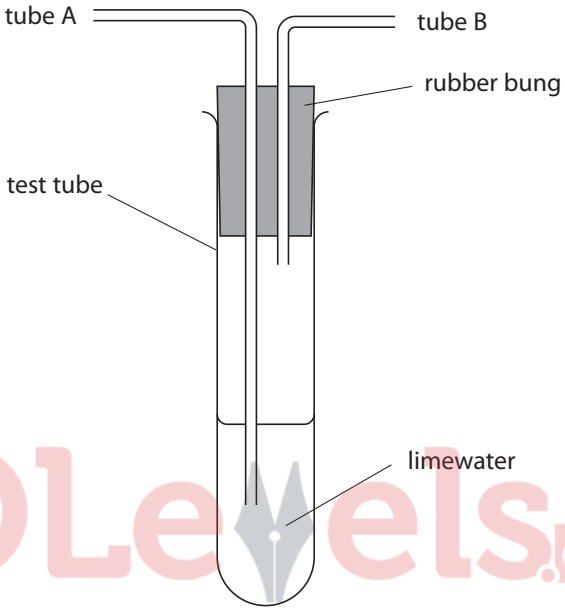
7. Question 5 · part a.i · 4 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

5 The amount of carbon dioxide in inhaled air is different from the amount of carbon dioxide in exhaled air.

The diagram shows a piece of apparatus that can be used to investigate this difference.



(a) (i) Describe how this apparatus should be used to compare the amount of carbon dioxide in inhaled and exhaled air.

(4)

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7. Question 5 · part a.i · 4 marks

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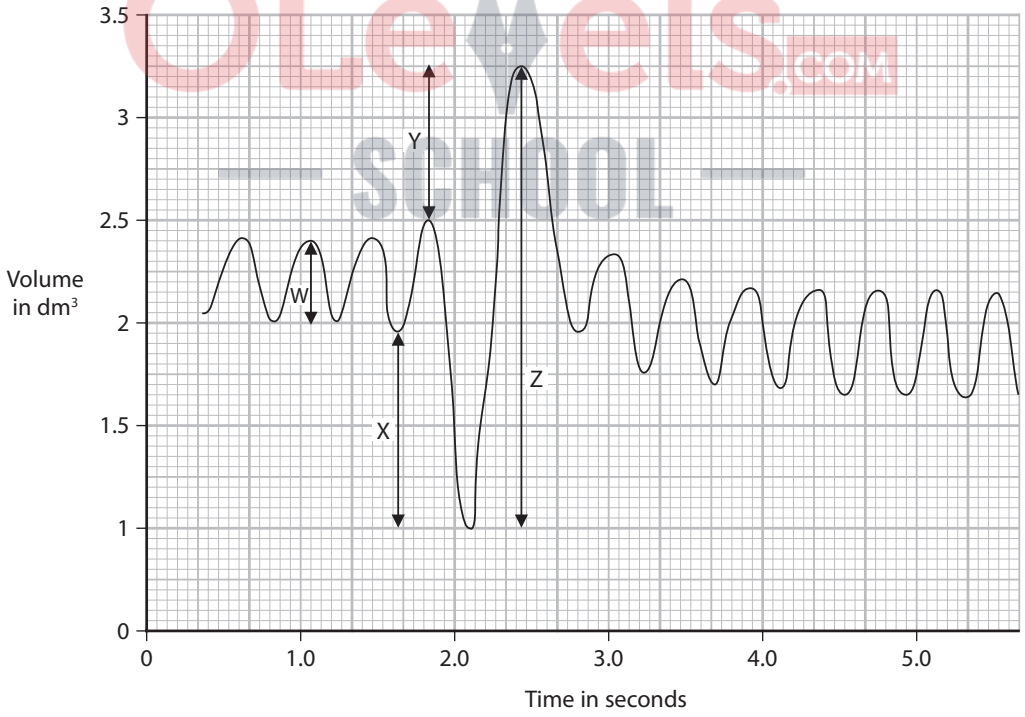
Further practice · Attempt part a.i only.

(ii) Explain the difference expected between the results.

(3)

(b) The movement of air in and out of the lungs can be measured using a spirometer.

The diagram shows a trace produced by a spirometer.



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7. Question 5 · part a.i · 4 marks · Working and answer

Working and answer

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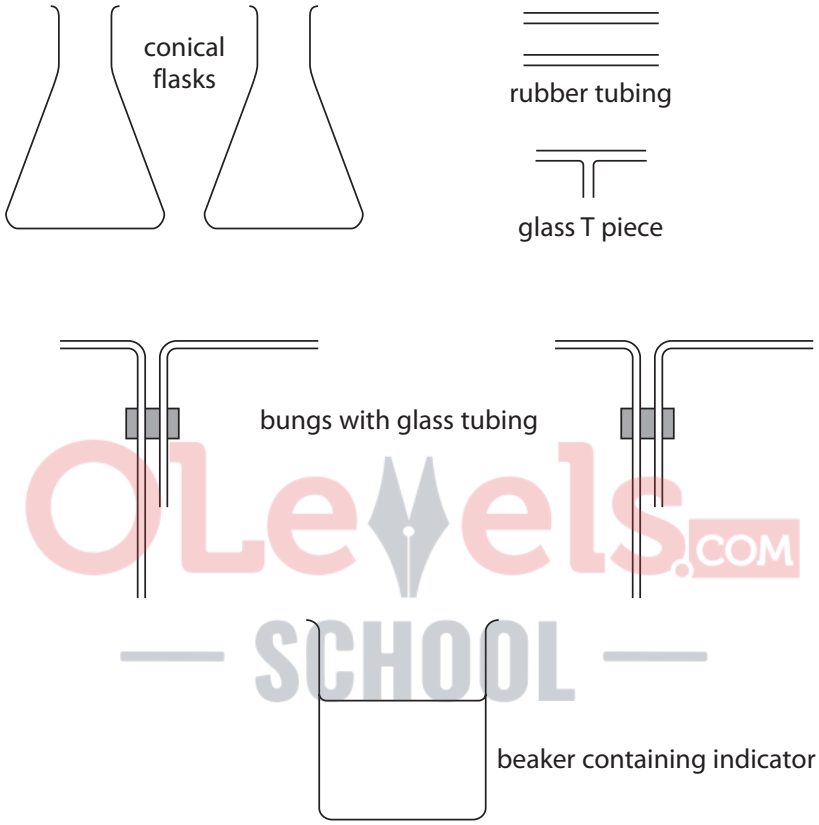
8. Question 1 · part c · 4 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

1 A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.



The apparatus is not drawn to scale.

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8. Question 1 · part c · 4 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

(3)

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(b) Name an indicator that could be used in the investigation.

(1)

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

(4)

8. Question 1 · part c · 4 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

(d) State one safety precaution.

(1)

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.....

(e) State one variable that should be controlled.

(1)

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(Total for Question 1 = 10 marks)

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Working and answer

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9. Question 7 · part a.iv · 4 marks

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Further practice · Attempt part a.iv only.

7 (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

(1)

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

(2)

$C_6H_{12}O_6$ + ... ⟶ ..... + ... CO_2

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

(2)

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9. Question 7 · part a.iv · 4 marks

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Further practice · Attempt part a.iv only.

(iv) Describe how a student could use this equipment and a timer to compare the rate of carbon dioxide production during exercise and at rest.

(4)

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9. Question 7 · part a.iv · 4 marks

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Further practice · Attempt part a.iv only.

(b) Aerobic and anaerobic respiration both produce ATP.

Describe the role of ATP in energy transfer during exercise.

(5)

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(Total for Question 7 = 14 marks)

9. Question 7 · part a.iv · 4 marks · Working and answer

Working and answer

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